Time Series Imputation — Algorithm Reference

Algorithm Metadata

60 algorithms across 6 categories, each with the paper it comes from and whether ImputeGAP implements it. Of the 60, 37 are integrated, 4 are in progress and 19 are absent, so a little under two thirds of the published work can be run from one interface. Citation counts are Google Scholar figures recorded in March 2026 and move over time, so read them as an indication of how much attention a paper has had.

Algorithm	Full Paper Name	Authors	Year	Venue	Cit.	ImputeGAP
Foundation Models						
NuwaTS	NuwaTS: A Foundation Model Mending Every Incomplete Time Series	Cheng et al.	2024	arXiv	10	yes
GPT4TS	One Fits All: Power General Time Series Analysis by Pretrained LM	Zhou et al.	2023	NIPS	1139	yes
MOMENT	MOMENT: A Family of Open Time-series Foundation Models	Goswami et al.	2024 2025	ICML / PMLR ICLR	636	*in progress*
Timer	Timer: Generative Pre-trained Transformers Are Large Time Series Models	Liu et al.	2024	ICML / PMLR	191	no
UniTS	UniTS: A Unified Multi-Task Time Series Model	Gao et al.	2024	NIPS	72	no
Deep Learning						
MissNet	Mining of Switching Sparse Networks for Missing Value Imputation in Multivariate Time Series	Obata, Kawabata, Matsubara, Sakurai	2024	KDD	11	yes
MPIN	Missing Value Imputation for Multi-attribute Sensor Data Streams via Message Propagation	Li, Li, Lu, Jensen, Pandey, Markl	2024	VLDB	18	yes
BitGraph	Biased Temporal Convolution Graph Network for Time Series Forecasting with Missing Values	Chen, Li, Liu, Li	2024	ICLR	39	yes
BayOTIDE	BayOTIDE: Bayesian Online Multivariate Time Series Imputation with Functional Decomposition	Fang, Wen, Luo, Zhe, Sun	2024	ICML / PMLR	27	yes
TimesNet	TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis	Wu, Hu, Liu, Zhou, Wang, Long	2023	ICLR	3459	yes
SAITS	SAITS: Self-attention-based Imputation for Time Series	Du, Côté, Liu	2023	ESWA	603	yes
PriSTI	PriSTI: A Conditional Diffusion Framework for Spatiotemporal Imputation	Liu, Huang, Feng, Sun, Du, Fu	2023	ICDE	154	yes
GRIN	Filling the Gaps: Multivariate Time Series Imputation by Graph Neural Networks	Cini, Marisca, Alippi	2022	ICLR	304	yes
CSDI	CSDI: Conditional Score-based Diffusion Models for Probabilistic Time Series Imputation	Tashiro, Song, Song, Ermon	2021	NIPS	1143	yes
DeepMVI	Missing Value Imputation on Multidimensional Time Series	Bansal, Deshpande, Sarawagi	2021	VLDB	116	yes
HKMF-T	HKMF-T: Recover From Blackouts in Tagged Time Series With Hankel Matrix Factorization	Wang, Wu, Wu, Tao, Lu	2021	TKDE	11	yes
SSGAN	Generative Semi-supervised Learning for Multivariate Time Series Imputation	Miao, Wu, Wang, Gao, Mao, Yin	2021	AAAI	254	*in progress*
GP-VAE	GP-VAE: Deep Probabilistic Multivariate Time Series Imputation	Fortuin, Baranchuk, Rätsch, Mandt	2020	AISTATS	461	*in progress*
NAOMI	NAOMI: Non-Autoregressive Multiresolution Sequence Imputation	Liu, Yu, Zheng, Zhan, Yue	2019	NIPS	233	*in progress*
M-RNN	Estimating Missing Data in Temporal Data Streams Using Multi-Directional Recurrent Neural Networks	Yoon, Zame, van der Schaar	2019	IEEE Trans. Biomed.	446	yes
BRITS	BRITS: Bidirectional Recurrent Imputation for Time Series	Cao, Wang, Li, Zhou, Li, Li	2018	NIPS	1203	yes
GAIN	GAIN: Missing Data Imputation using Generative Adversarial Nets	Yoon, Jordon, van der Schaar	2018	ICML / PMLR	2014	yes
WGAN	Multivariate Time Series Imputation with Generative Adversarial Networks	Luo et al.	2018	NIPS	736	no
E^2GAN	E^2GAN: End-to-End Generative Adversarial Network for Multivariate Time Series Imputation	Luo et al.	2019	IJCAI	422	no
SSSD	Diffusion-based Time Series Imputation and Forecasting with Structured State Space Models	Alcaraz, Strodthoff	2023	TMLR	331	no
MTSIT	Multivariate Time Series Imputation With Transformers	Yıldız et al.	2022	IEEE SPL	106	no
TIDER	Multivariate Time-series Imputation with Disentangled Temporal Representations	Li et al.	2023	ICLR	55	no
PoGeVon	Networked Time Series Imputation via Position-aware Graph Enhanced Variational Autoencoders	Wang et al.	2023	KDD	49	no

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CSBI	Provably Convergent Schrödinger Bridge with Applications to Probabilistic Time Series Imputation	Chen et al.	2023	ICML / PMLR	48	no
TimeCIB	Conditional Information Bottleneck Approach for Time Series Imputation	Choi et al.	2024	ICLR	23	no
FGTI	Frequency-aware Generative Models for Multivariate Time Series Imputation	Yang et al.	2024	NIPS	20	no
CATSI	Context-aware Time Series Imputation for Multi-analyte Clinical Data	Yin et al.	2020	JHIR	15	no
CSAI	CSAI: Conditional Self-Attention Imputation for Healthcare Time-series	Qian et al.	2026	arXiv	3	no
SST-GCN	Spatio-temporal Graph Convolutional Network for Stochastic Traffic Speed Imputation	Muniz Cuza et al.	2022	SIGSPATIAL	6	no
FIM-ℓ	Zero-shot Imputation with Foundation Inference Models for Dynamical Systems	Seifner et al.	2025	ICLR	6	no
ReCTSi	ReCTSi: Resource-efficient Correlated Time Series Imputation via Decoupled Pattern Learning	Lai et al.	2024	KDD	5	no
SSD-TS	SSD-TS: Exploring the Potential of Linear State Space Models for Diffusion Models in Time Series Imputation	Gao et al.	2025	KDD	8	no
LSCD	LSCD: Lomb-Scargle Conditioned Diffusion for Time Series Imputation	Fons et al.	2025	ICML / PMLR	1	no
DMM	Causal View of Time Series Imputation: Some Identification Results on Missing Mechanism	Cai et al.	2025	IJCAI	0	no
Matrix Completion						
CDRec	Scalable Recovery of Missing Blocks in Time Series with High and Low Cross-correlations	Khayati, Cudré-Mauroux, Böhlen	2020	KAIS	22	yes
TRMF	Temporal Regularized Matrix Factorization for High-dimensional Time Series Prediction	Yu, Rao, Dhillon	2016	NIPS	746	yes
GROUSE	Global Convergence of a Grassmannian Gradient Descent Algorithm for Subspace Estimation	Zhang, Balzano	2016	AISTATS / PMLR	82	yes
ROSL	Robust Orthonormal Subspace Learning: Efficient Recovery of Corrupted Low-Rank Matrices	Shu, Porikli, Ahuja	2014	CVPR	130	yes
SoftImpute	Spectral Regularization Algorithms for Learning Large Incomplete Matrices	Mazumder, Hastie, Tibshirani	2010	JMLR	2058	yes
SVT	A Singular Value Thresholding Algorithm for Matrix Completion	Cai, Candès, Shen	2010	SIAM J. OPTIM	7110	yes
SPIRIT	Streaming Pattern Discovery in Multiple Time-Series	Papadimitriou, Sun, Faloutsos	2005	VLDB	582	yes
IterativeSVD	Missing Value Estimation Methods for DNA Microarrays	Troyanskaya et al.	2001	Bioinformatics	5446	yes
Pattern Search						
TKCM	Continuous Imputation of Missing Values in Streams of Pattern-Determining Time Series	Wellenzohn, Böhlen, Dignös, Gamper, Mitterer	2017	EDBT	44	yes
ST-MVL	ST-MVL: Filling Missing Values in Geo-Sensory Time Series Data	Yi, Zheng, Zhang, Li	2016	IJCAI	255	yes
DynaMMo	DynaMMo: Mining and Summarization of Coevolving Sequences with Missing Values	Li, McCann, Pollard, Faloutsos	2009	KDD	226	yes
Machine Learning						
IIM	Learning Individual Models for Imputation	Zhang, Song, Sun, Wang	2019	ICDE	67	yes
XGBoost	XGBoost: A Scalable Tree Boosting System	Chen, Guestrin	2016	KDD	74297	yes
MICE	Multiple Imputation by Chained Equations (MICE): Implementation in Stata	Royston, White	2011	J. Stat. Software	1518	yes
MissForest	MissForest: Non-parametric Missing Value Imputation for Mixed-type Data	Stekhoven, Bühlmann	2012	Bioinformatics	6952	yes
Statistics						
KNNImpute	K-Nearest Neighbour Imputation (classical method)	—	—	—	—	yes
Interpolation	Linear / Spline Interpolation (classical method)	—	—	—	—	yes
MinImpute	Min-value Imputation (classical method)	—	—	—	—	yes
ZeroImpute	Zero Imputation (classical method)	—	—	—	—	yes
MeanImpute	Mean Imputation (classical method)	—	—	—	—	yes
MeanImputeBy-Series	Per-series Mean Imputation (classical method)	—	—	—	—	yes